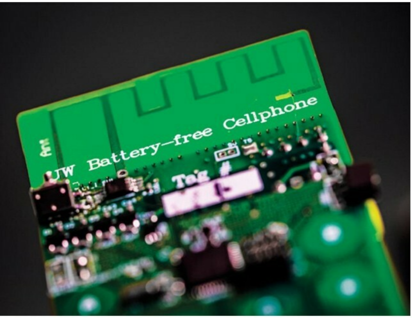
World's first battery-free cellphone developed

University of Washington researchers have developed a cellphone that requires no batteries—a major leap forward in moving beyond chargers, cords and dying phones. Instead, the phone is powered from either ambient radio signals or light. The team made Skype calls using their battery-free phone, demonstrating that the prototype made of commercial, off-the-shelf components can receive and transmit speech and communicate with a base station.



The battery-free phone developed at the University of Washington can sense speech, actuate the earphones, and switch between uplink and downlink communications, all in real time. It is powered by either ambient radio signals or light (Image courtesy: Mark Stone/University of Washington)

To achieve this feat, researchers eliminated a power-hungry step in most modern cellular transmissions—converting analogue signals that convey sound into digital data that a phone can understand. This process consumes so much energy that it's been impossible to design a phone that can rely on ambient power sources.

The battery-free cellphone takes advantage of tiny vibrations in a phone's microphone or speaker that occur when a person is talking into a phone or listening to a call. An antenna connected to these components converts their motion into changes in standard analogue radio signal emitted by a cellular base station.

To transmit speech, the phone uses vibrations from the device's microphone to encode speech patterns in the reflected signals. To receive speech, it converts encoded radio signals into sound vibrations, which are picked up by the phone's speaker. In the prototype device, the user presses a button to switch between 'transmitting' and 'listening' modes.

Robot driver that makes any car autonomous

A robotic driver is on its way to relieve us from the stress of the daily commute. Once installed in a vehicle, which takes only a few minutes, it would free us to do more important tasks or just sit back and relax.

The robot, called IVO for 'intelligent vehicle operator,' consists of a combination of vision sensors and mechanical systems to steer and operate the gas and brake pedals. Users input a location using Google Maps and the robot drives there autonomously while detecting and avoiding obstacles.

Conceived by a team led by Hugo Guterman, a professor at Ben-Gurion University of the Negev, the portable IVO works like a human driver. Users belt it into the driver's seat. A three-pronged gripper at the end of an

IVO driver robot (Image courtesy: Ben-Gurion University of the Negev)